

Dinupa Devinda

Machine Learning Focused Graduate | AI/ML Developer | R&D Engineer
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PROFILE

Machine Learning focused engineering graduate with a background in computer science, mathematics, Electrical and Electronics Engineering, and data analysis. Experienced in building practical ML systems, from data preparation and model development to evaluation and deployment focused implementation. Interested in developing intelligent, data driven solutions for real world technology and engineering problems.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)
BSc. Engineering (Hons) in Electrical and Electronic Engineering
2022 – 2026 | Sri Lanka

University of Kelaniya
BSc. in Physical Science
(Subjects: Computer Science, Pure Mathematics, Applied Mathematics)
2022 – 2026 | Sri Lanka

SKILLS & EXPERTISE

Programming Languages: Python | C/C++ | Java | JavaScript | C# | SQL

Frameworks & Tools: PyTorch | TensorFlow/Keras | scikit-learn | XGBoost | Optuna | NumPy | Pandas | OpenCV | YOLO • ONNX Runtime | TensorFlow Lite | NCNN | Spring Boot | React | MongoDB | SQL Server | Git | GitHub Actions | MQTT/MQTTS | ThingsBoard

Domains: Machine Learning | Deep Learning | Computer Vision | Edge AI | Sensor Fusion | Audio Classification | Time-Series Classification | Data Analysis

PROFESSIONAL EXPERIENCE

Research and Development Engineering Intern

Sri Lanka Telecom – Digital Lab | Hybrid | June 2025 – September 2025

- Developed C/C++ firmware for ESP32-based edge devices with A7680C 4G modem AT-command handling, MQTT/MQTTS, HTTP, SMTP, SMS, and call support.
- Built device-to-cloud telemetry pipelines using MQTT/HTTP and structured JSON payloads for live monitoring, alerts, remote status reporting, and downstream analysis.
- Configured ThingsBoard dashboards and custom web portals, using MQTT Explorer, HiveMQ, and basic Grafana exposure for topic testing, JSON payload validation, telemetry debugging, and basic GitHub Actions/CI workflow practice.

Research and Development Inplant Trainee – Electronic Manufacturing Services Floor

Variosystems (Pvt) Ltd | Onsite | June 2024 – September 2024

- Built C/C++ based automation and embedded-control projects for a manufacturing environment, including an Automated Guided Vehicle, flame-treatment automation upgrade, and automated PCB conveyor system.
- Gained hands-on exposure to production data flow, SMT/THT manufacturing, ICT and functional testing, SPEA 3030/4050 test systems, FactoryLogix workflows, and NPI processes.

PROJECTS

Vehicular Black Box: Vehicle Violation Detector

Machine Learning, Edge AI, Sensor Fusion| Nov 2025 – May 2026

- Built and integrated a Raspberry Pi based vehicle violation detection system combining computer vision, audio classification, IMU-based ML, GPS/GSM communication, backend APIs, dashboard visualization, and driver scoring, with documented datasets, model cards, evaluation summaries, and a BYD Atto BMI160 IMU dataset published on [Zenodo/Kaggle](#).

Road Sign Detection: Developed a two-stage computer vision inference pipeline using YOLO, NCNN, ONNX Runtime, and OpenCV for Raspberry Pi deployment, evaluated on 9,062 test images.

Horn Detection: Built an audio classification model using log-mel spectrogram feature extraction, TensorFlow/Keras CNN training, and TensorFlow Lite deployment for live microphone inference.

Crash Detection: Developed crash-detection models using PyTorch CNN audio inference and TensorFlow/Keras CNN-GRU IMU time-series inference for multi-signal event detection.

Lane Change Detection: Built an IMU time-series classification model using accelerometer/gyroscope windowing, feature engineering, XGBoost training, and joblib deployment artifacts.

Aggressive Driving Detection: Developed an IMU-based machine learning classifier using yaw rate and raw acceleration windows, model comparison, Optuna-tuned XGBoost, threshold tuning, and evaluation.

ServiceLink: On-Demand Local Hiring Platform

Full-Stack Development, Spring Boot REST APIs| Oct 2025 – Jan 2026

- Built a full-stack service marketplace using Java 17, Spring Boot REST APIs, Spring Security/JWT authentication, MongoDB, React, Vite, React Router, React Query, and Axios, implementing role-based access, service listings, bookings, admin workflows, and API-driven frontend/backend communication.

A7680C GSM 4G Library for ESP32

C/C++, IoT Communication, MQTT/MQTTS, HTTP| June 2025 – Sep 2025

- Built a C++ ESP32 library for A7680C 4G modem AT-command handling, supporting modem initialization, APN/CSQ/session setup, MQTT/MQTTS over TLS, HTTP GET/POST, SMS/call features, SMTP email alerts, TLS certificate configuration, and web-based device configuration.

Industrial Machine Monitoring System with Dashboard

Edge Telemetry, IoT Data Pipelines, MQTT/TLS, ThingsBoard| June 2025 – Sept 2025

- Built ESP32 and A7680C based industrial telemetry systems that collected machine data, cross-checked readings against reference instruments, and streamed secure MQTTS JSON payloads to ThingsBoard and browser-based WebSocket MQTT dashboards with connection health checks, alerts, remote monitoring, and MQTT topic debugging.

Other Projects

Automation Systems, Backend APIs, Embedded C/C++, Database Applications

- Built automation and monitoring projects including a Raspberry Pi based **Smart Fish Habitat Management System**, an Arduino based 100 Kg **Automated Guided Vehicle**, and an **automated PCB conveyor system**, applying Python, embedded C/C++, control logic, real-time monitoring, process automation, sensor data handling, and system integration.; completed additional projects including **BRMS Boarding Room Management** API with Java, Spring Boot REST APIs, Spring Data JPA, SQL Server, environment-based configuration, and H2-backed testing, Music Streaming Management System in Java, **Public Transport file-based database system** in C, **Music Player App** in C#/.NET, **EduPress Store** using PHP/MySQL/XAMPP, DS1302 RTC **assembly mini project**.

CERTIFICATIONS

- PyTorch Bootcamp (OpenCV University, 2025)
- AI/ML Engineer Stage 2 (SLIIT, 2025)
- AI/ML Engineer Stage 1 (SLIIT, 2023)
- Python for Beginners (University of Moratuwa, 2023)
- IT Essentials (Cisco, 2025)
- Microcontroller Programming: Embedded C and Assembly Language (MindLuster, 2024)
- Certificate in Business Accounting (CIMA, 2023)